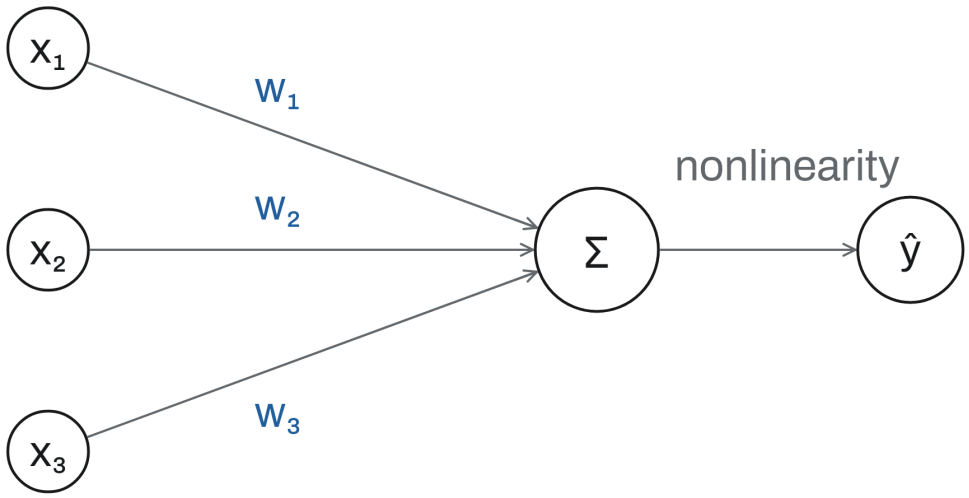
Deep learning architectures

Qihong Lu

Notes

The same ingredients



Inputs contribute through weights.

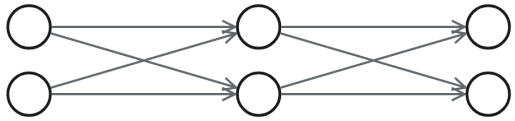
A nonlinearity turns the sum into an output.

Training changes the weights to reduce prediction error.

Notes

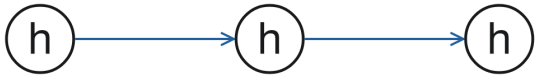
Architecture changes the information flow

Layers



Layers combine features.

Recurrence



Recurrence carries information through time.

Convolution

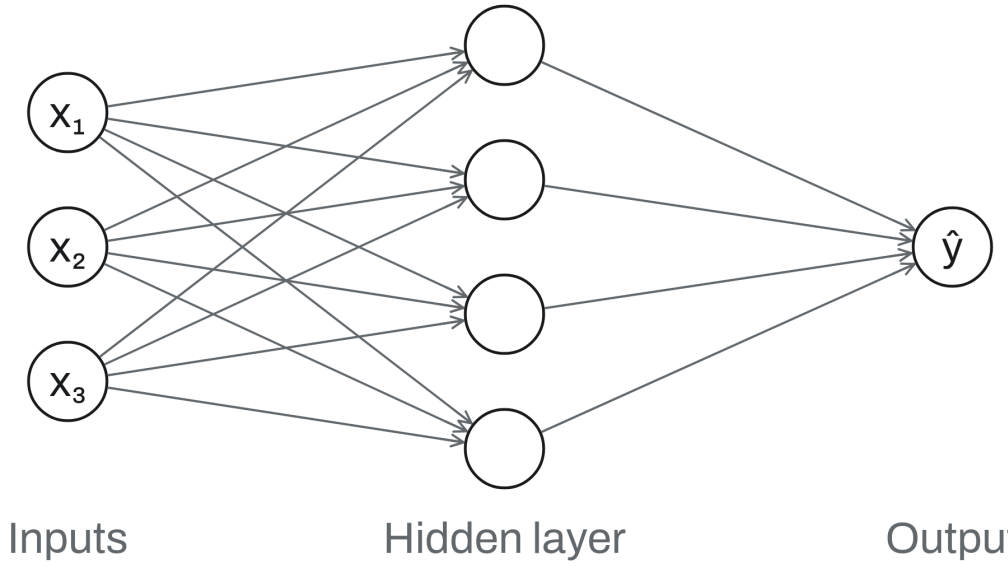
Convolution reuses a local pattern.

Attention

Attention mixes information from context.

Notes

A hidden layer creates a representation



A hidden layer transforms the inputs.

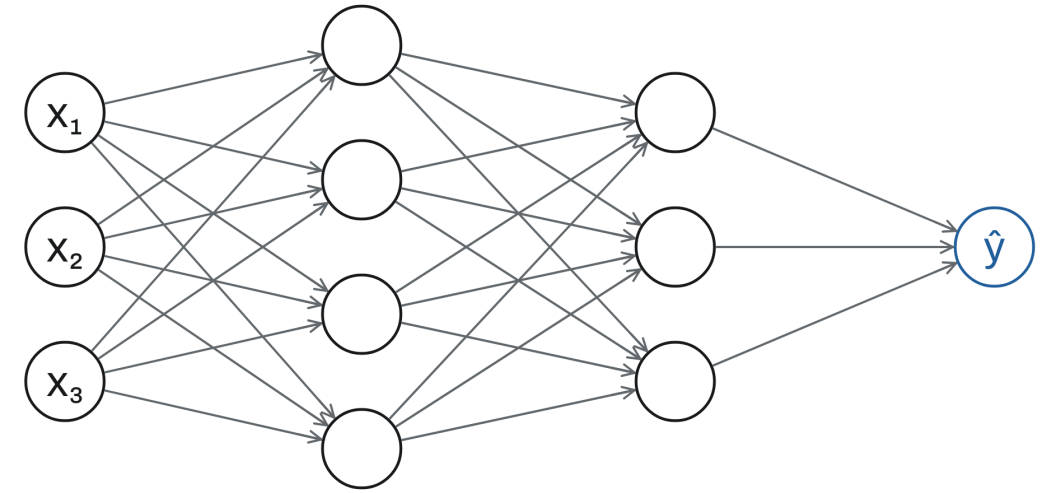
The output uses that transformed pattern.

If we freeze the hidden layer, what can training still change?

The output weights and bias can change. The hidden representation stays fixed.

Notes

A multilayer perceptron (MLP)

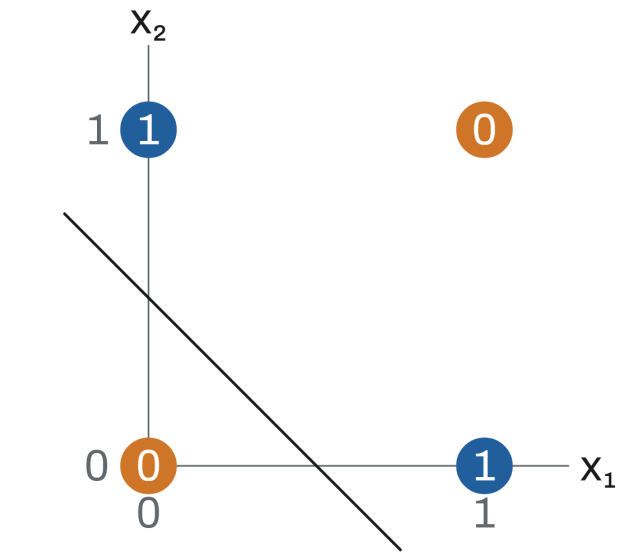


The next layer combines features from the previous layer.

Depth means repeated transformation.

Notes

XOR: one straight boundary is not enough



0: same inputs 1: different inputs

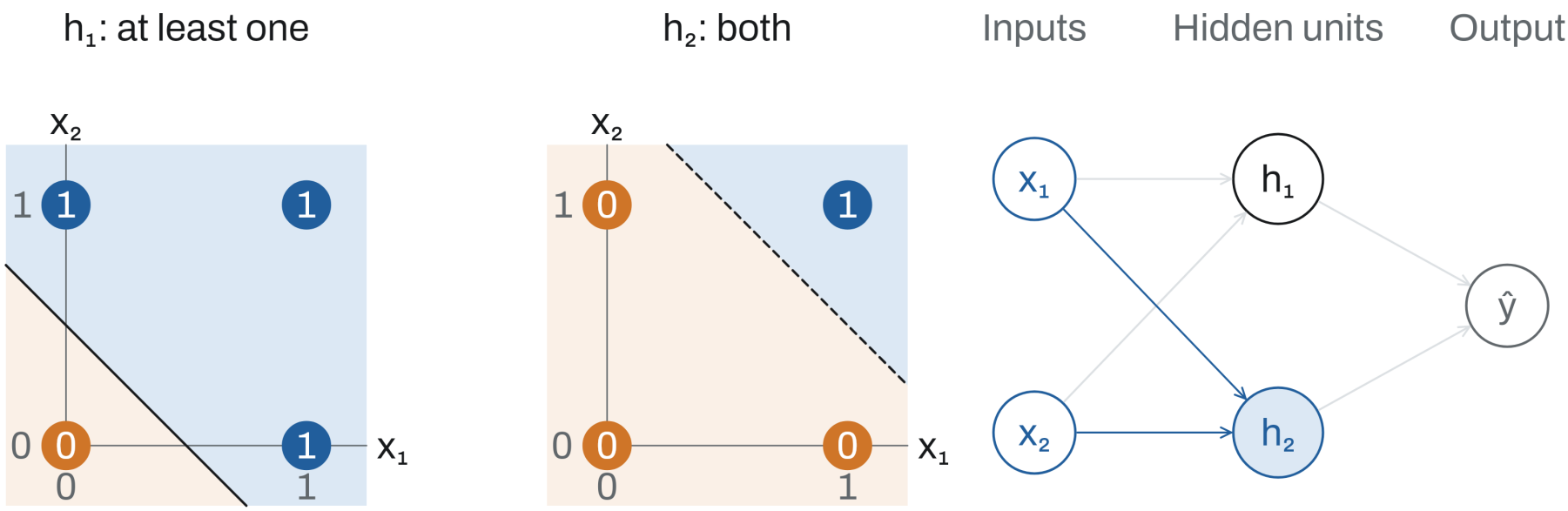
Output 1 when exactly one input is 1.

This straight boundary gets three of the four cases right.

Moving the line cannot separate the two diagonals.

Notes

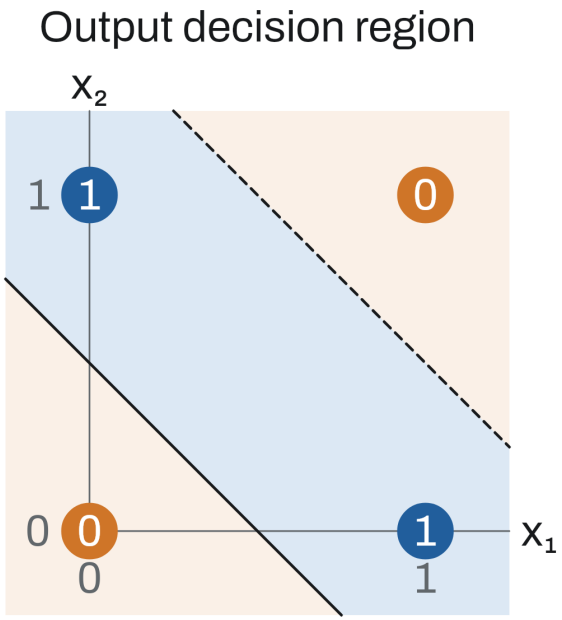
Two hidden units ask simpler questions

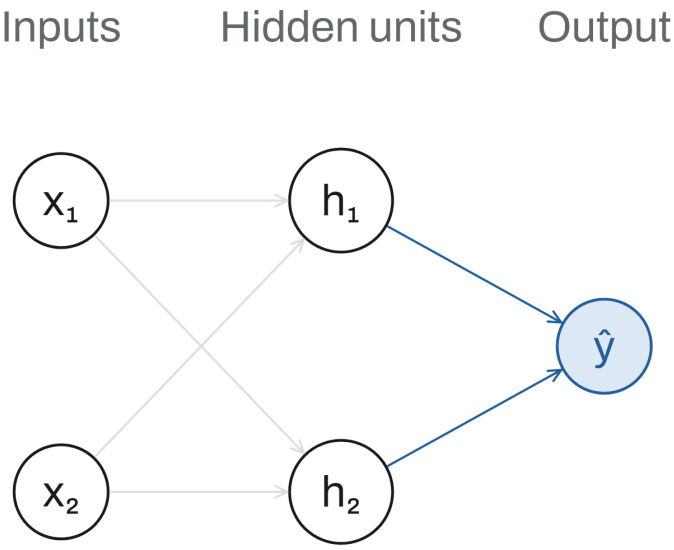


Step-function hidden units · colors show each unit's output

Notes

Combine the hidden decisions to solve XOR



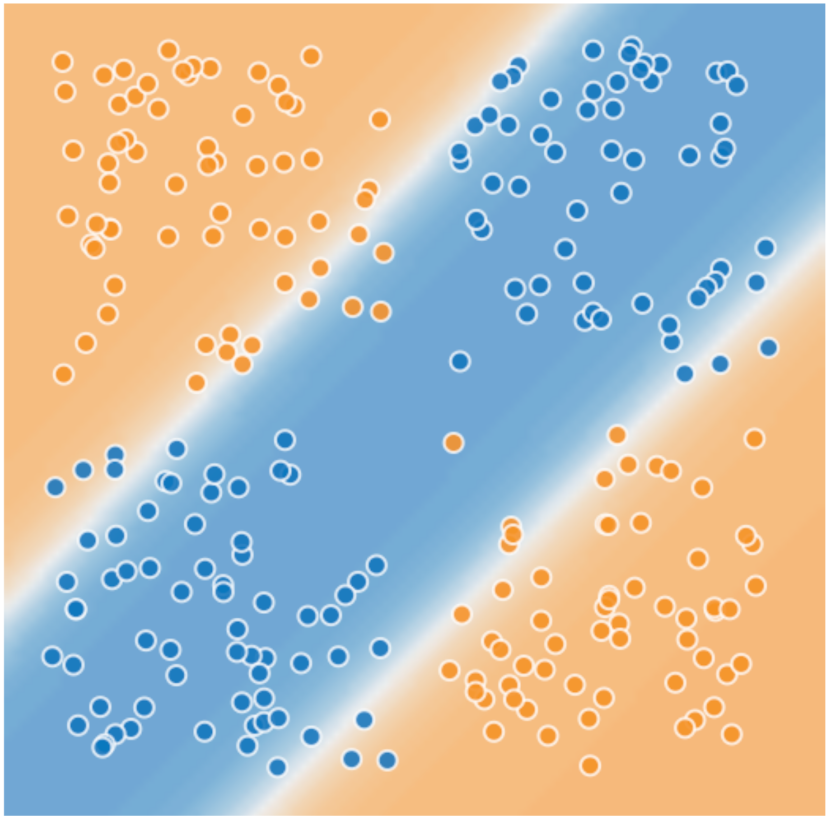


0: same inputs 1: different inputs

Exactly one input → output 1.

Notes

Learn this decision boundary with two hidden units



Neural Network Playground

Notes

Semantic memory

robin

has feathers
can fly

canary

has feathers
can fly

rose

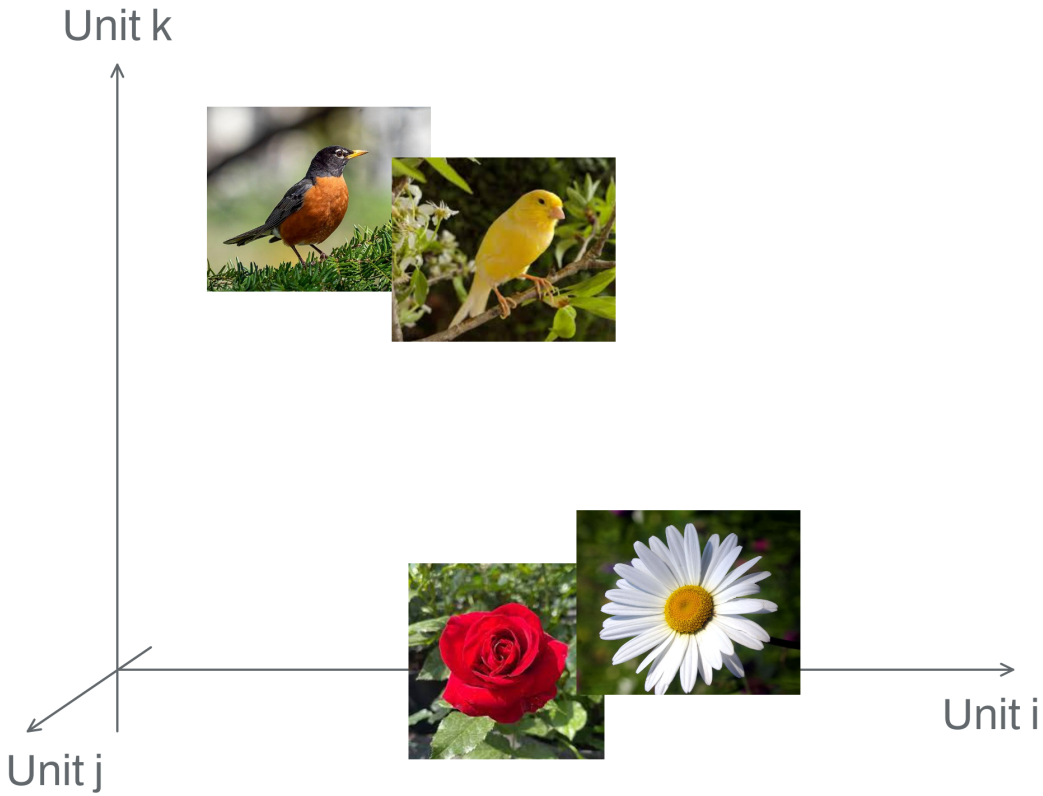
has petals
can grow

daisy

has petals
can grow

Notes

Concepts have a similarity structure



Similar concepts evoke similar neural patterns.

Notes

Concepts have a similarity structure

New Songbird Species Discovered in Bolivia

Jan 5, 2026 by Natali Anderson

« Previous | Next »

Published in
Biology
Featured

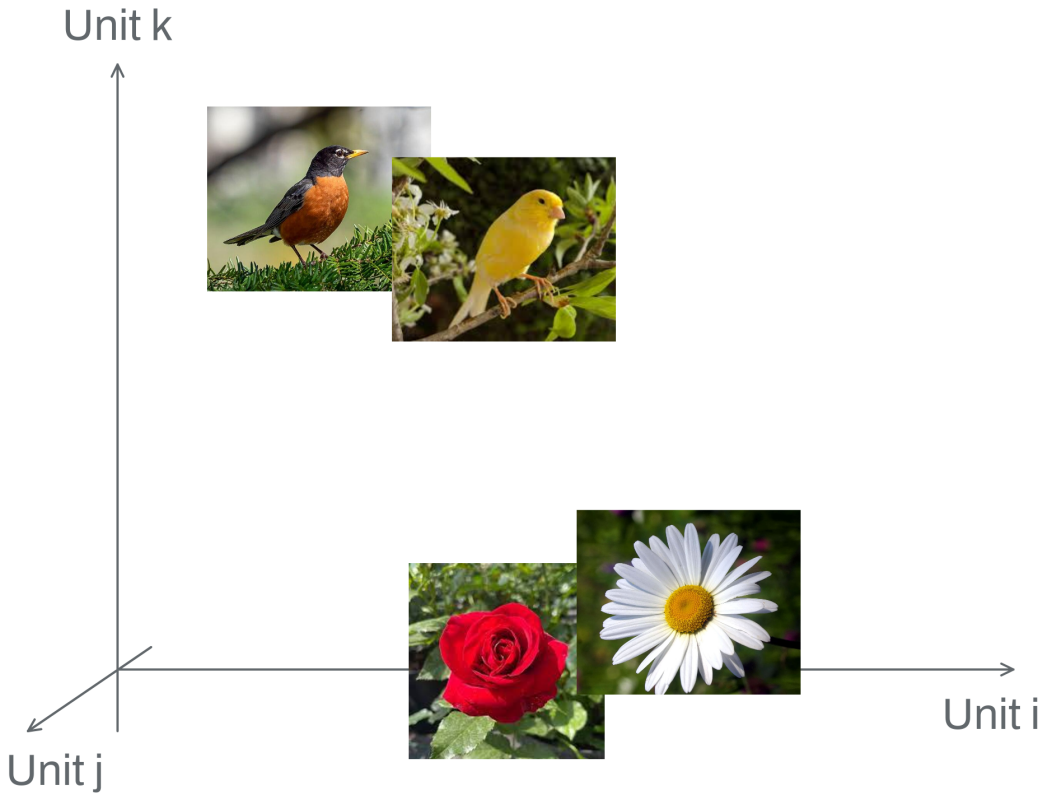
Tagged as
Beni greenlet
Bird
Bolivia
DNA
Gene

Deep in the seasonally flooded savannas of Bolivia, a small, olive-green songbird has been quietly eluding science. Now, after six decades of confusion and misidentification, ornithologists have confirmed that the bird is not just a regional variant of a known species in the greenlet genus *Hylophilus* but an entirely new one. The discovery adds a new endemic species to South America's already rich avian diversity and underscores how much remains unknown even in well-studied bird families.

Similar concepts evoke similar neural patterns.

Notes

Broad and fine semantic distinctions



Development: broad categories can emerge before finer distinctions.

Mandler & McDonough (1993)

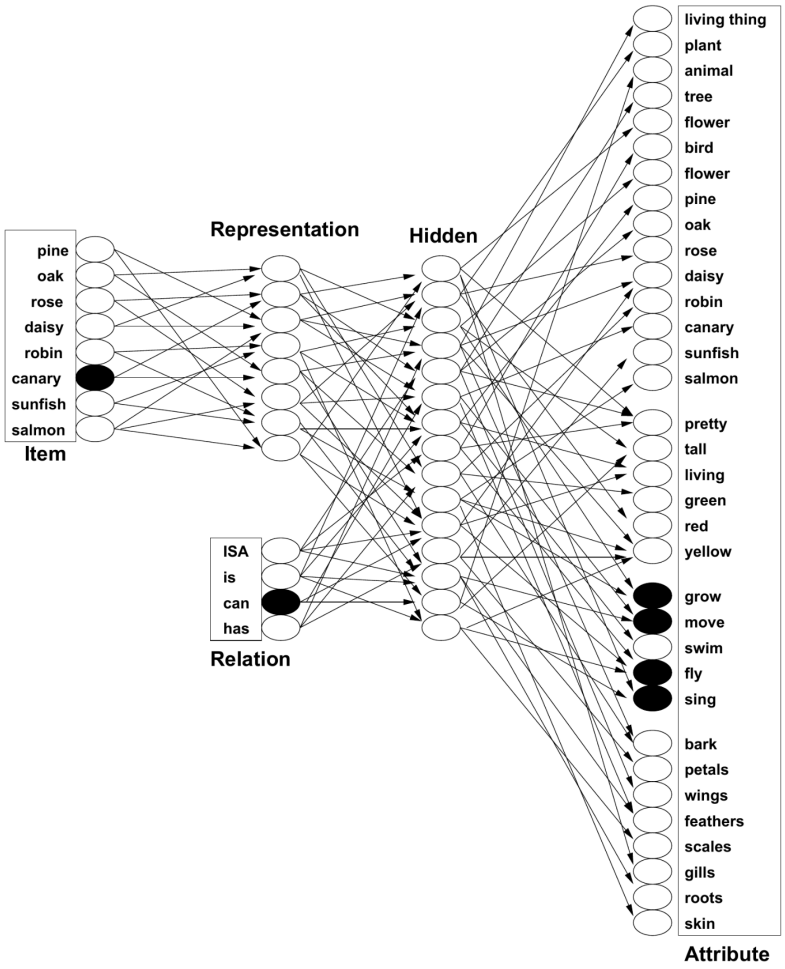
Semantic dementia: distinctive knowledge can be more vulnerable.

Rogers et al. (2004)

What learning process could produce this structure?

Notes

Semantic knowledge as a prediction task



Eight item inputs and four relation inputs.

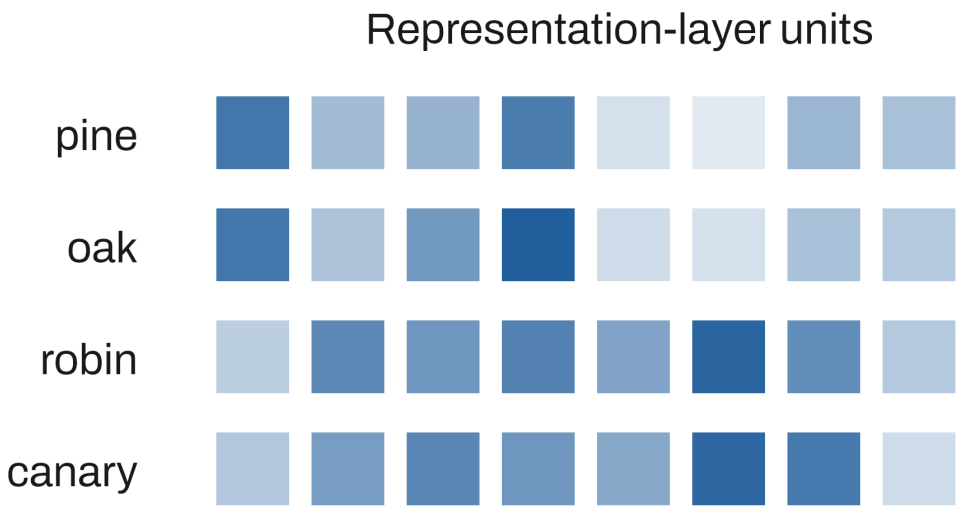
One item and one relation are selected for each query.

“Canary” + “can” → grow, move, fly, sing.

Rogers & McClelland (2004), Semantic Cognition · Chs. 2–3 · Figure 2.2

Notes

Read a distributed representation



Epoch 150 approximation · darker = more activity

Each column is a Representation-layer unit.

Each row shows the unit activations for one item.

Similar patterns can support similar predictions.

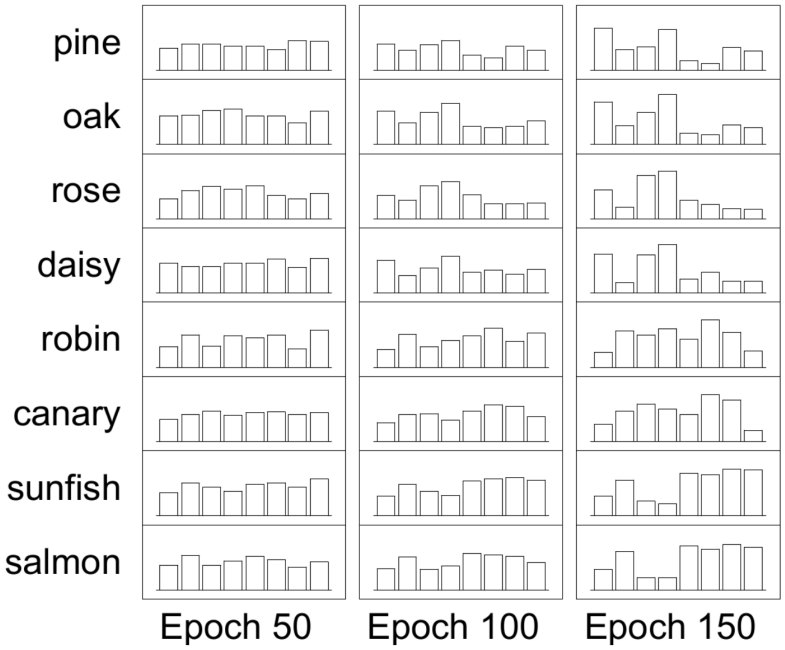
Does a similar pattern require one “plant neuron”?

No. Similarity can be distributed across several units.

Rogers & McClelland (2004), Semantic Cognition · Chs. 2–3 · Figure 3.1 · epoch 150 (approximation)

Notes

Learning separates semantic structure



Representation layer: eight learned units.

Each bar is one unit's activation for the item in its row.

Columns: training epochs 50, 100 and 150.

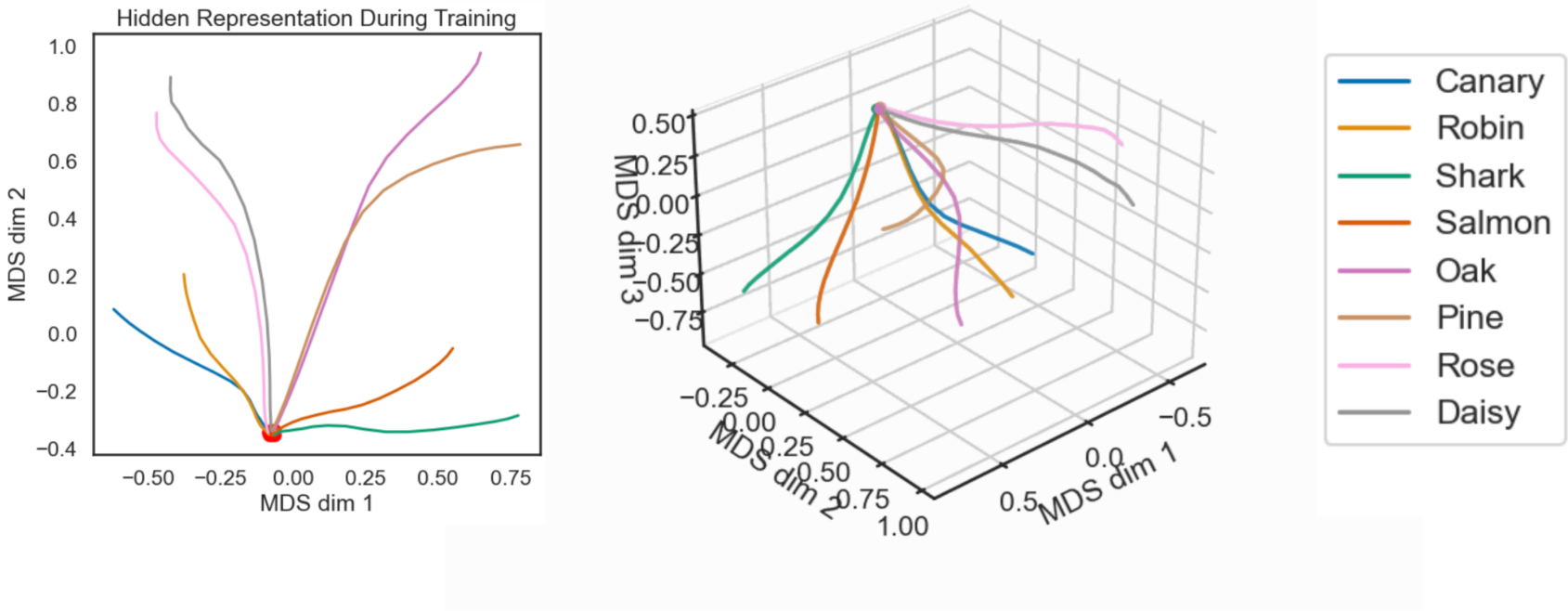
Related items develop similar activation patterns.

The task and training experience shape this semantic structure.

Rogers & McClelland (2004), Semantic Cognition · Chs. 2–3

Notes

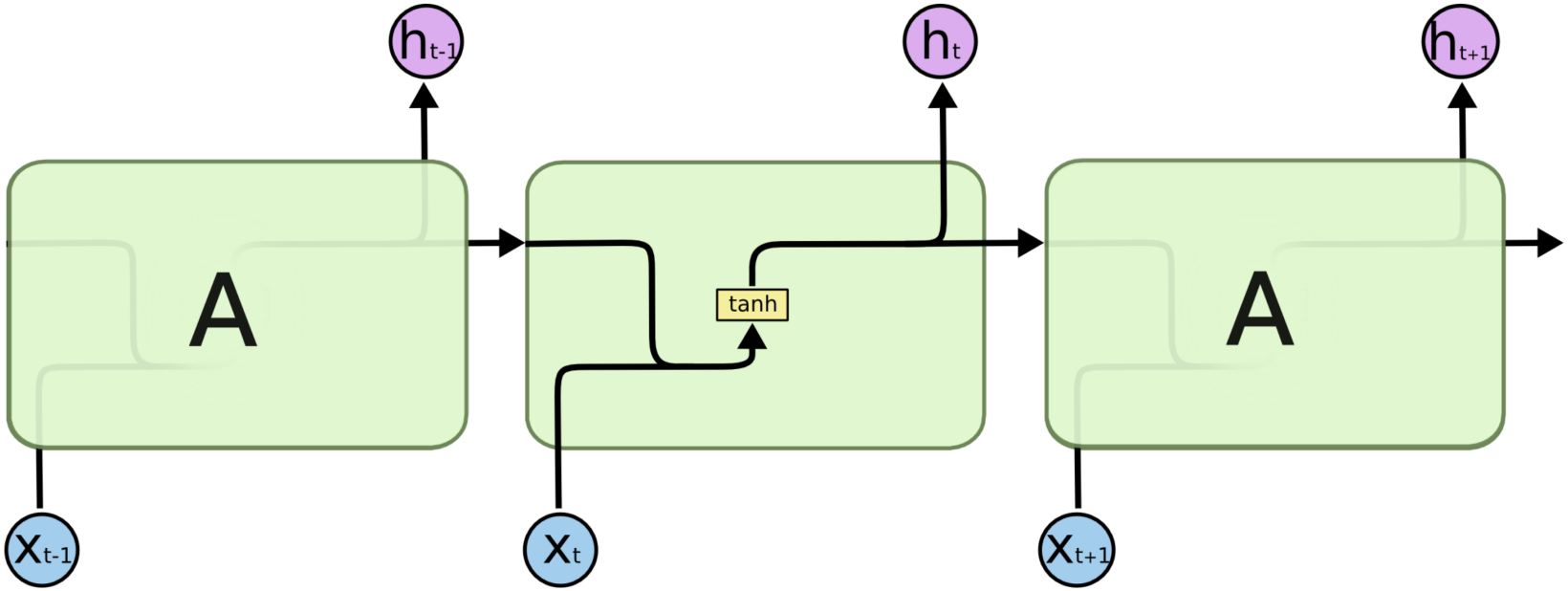
Learning differentiates concepts gradually



github.com/qihongl/demo-rumelhart-net

Notes

Recurrent neural network (RNN)



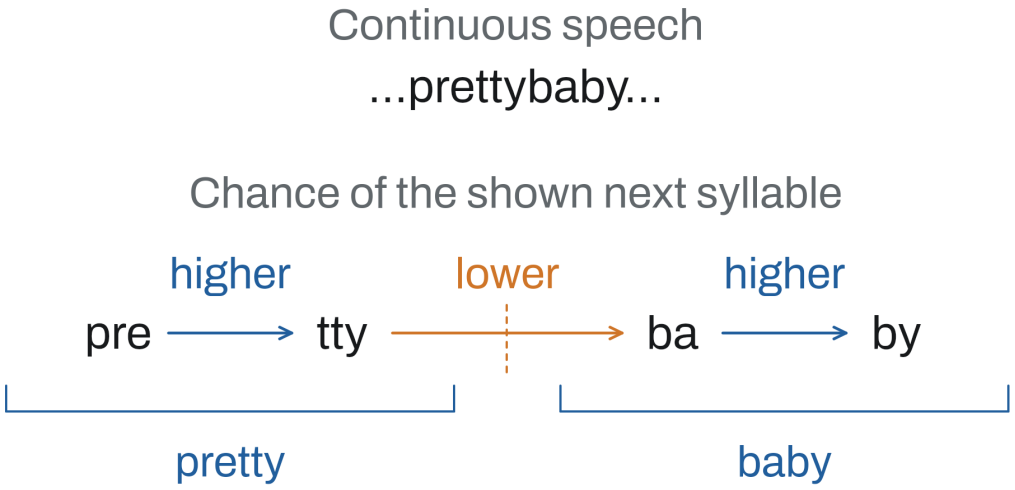
The new hidden state uses the input and the old hidden state.

The state carries information from earlier inputs.

Christopher Olah (2015) · Understanding LSTM Networks

Notes

How do we recognize word boundaries?



A drop in predictability can signal a boundary

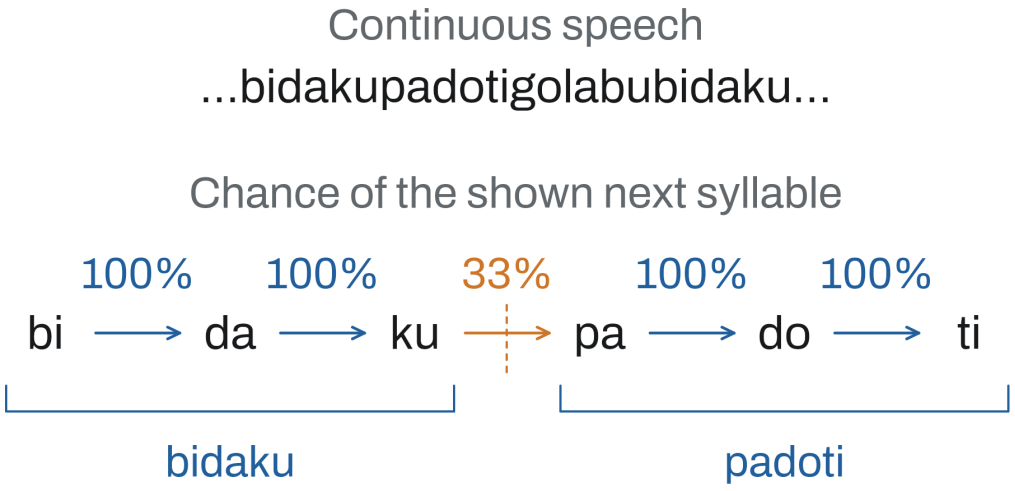
"pre" → "tty": higher transition probability.

"tty" → "ba": lower transition probability.

Saffran, Aslin & Newport (1996) · "pretty baby" example

Notes

Finding word boundaries without pauses



A drop in predictability can signal a boundary

8-month-old infants heard two minutes of continuous artificial speech.

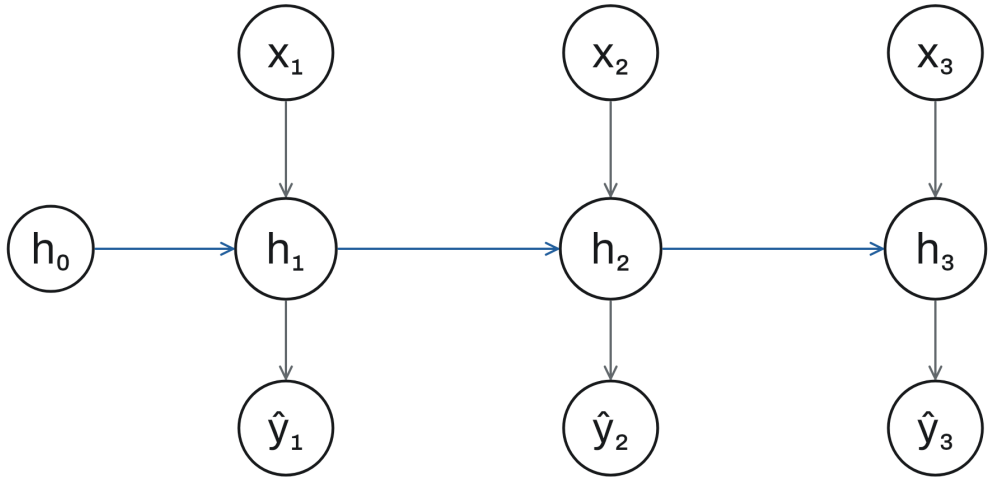
Syllables within a word were more predictable than syllables across words.

At test, infants distinguished words from sequences crossing a boundary.

Saffran, Aslin & Newport (1996) · Statistical Learning by 8-Month-Old Infants

Notes

Unroll the network through time



Shared weights across time

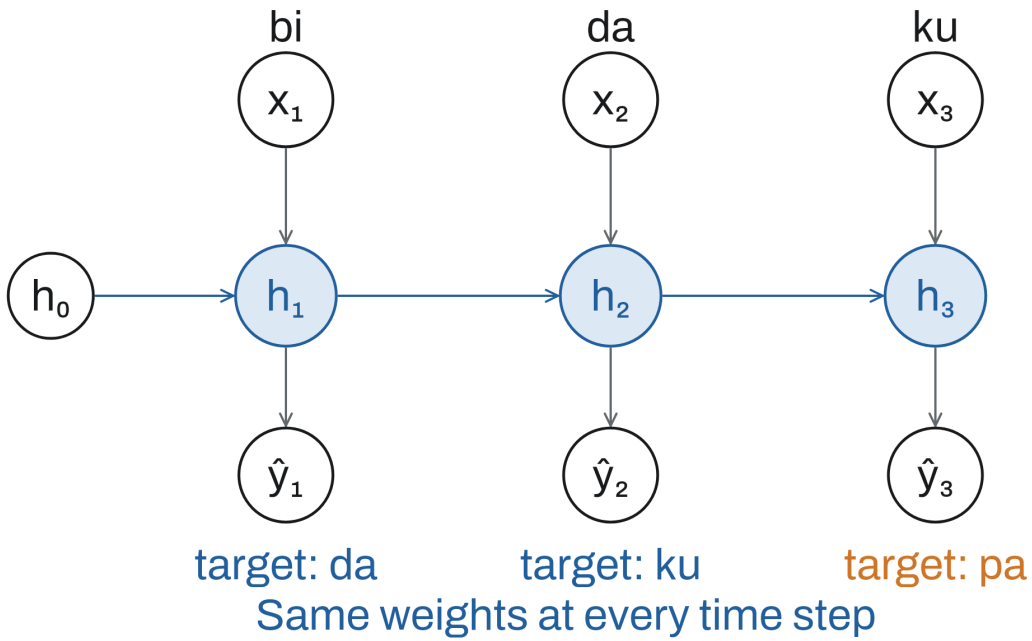
One recurrent network, shown at several moments.

The same weights are reused at every step.

Elman (1990) · architecture; Saffran, Aslin & Newport (1996) · syllables

Notes

Elman network: predict the next syllable



Example stream: … bi da ku pa do ti …

Training target: the next observed syllable.

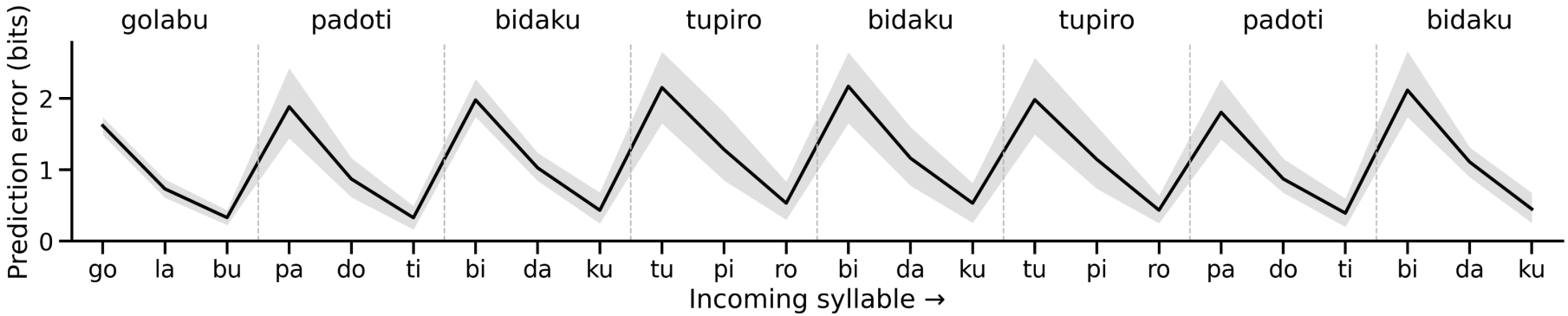
The hidden state carries earlier context.

Elman (1990) · architecture; Saffran, Aslin & Newport (1996) · syllables

Notes

Prediction error rises at word boundaries

One training exposure · 180 words / 540 syllables

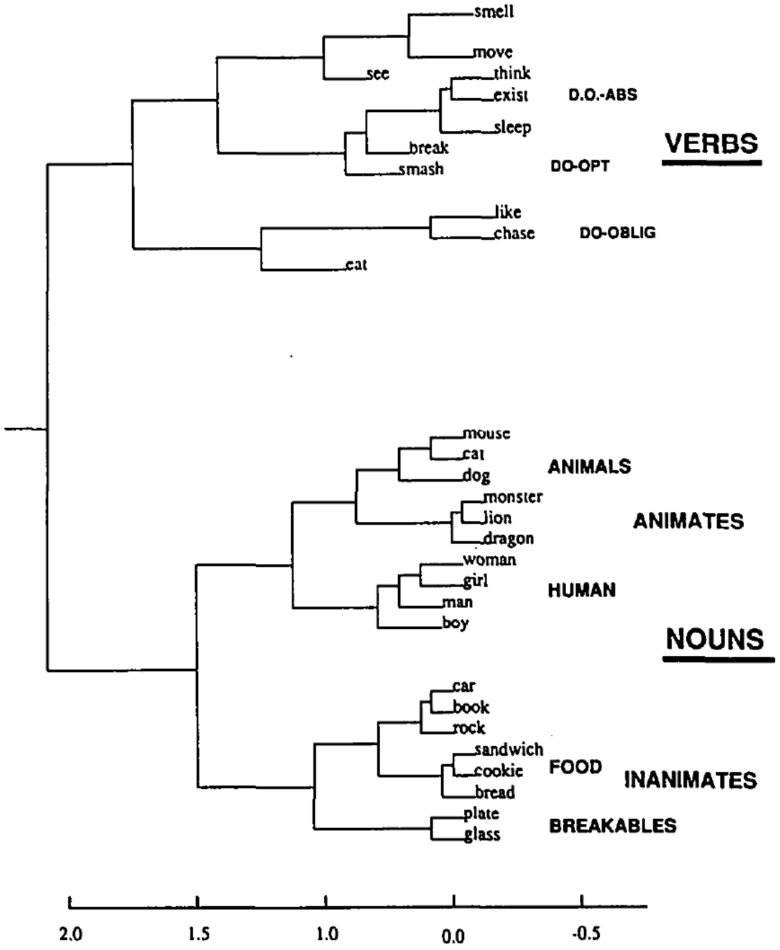


Mean of 5 models · gray band: 95% bootstrap CI

github.com/qihongl/demo-elman-1990

Notes

Structure appears in hidden states



Elman's original sentence experiment: words cluster by hidden activity.

Each word is represented by an average pattern across 150 hidden units.

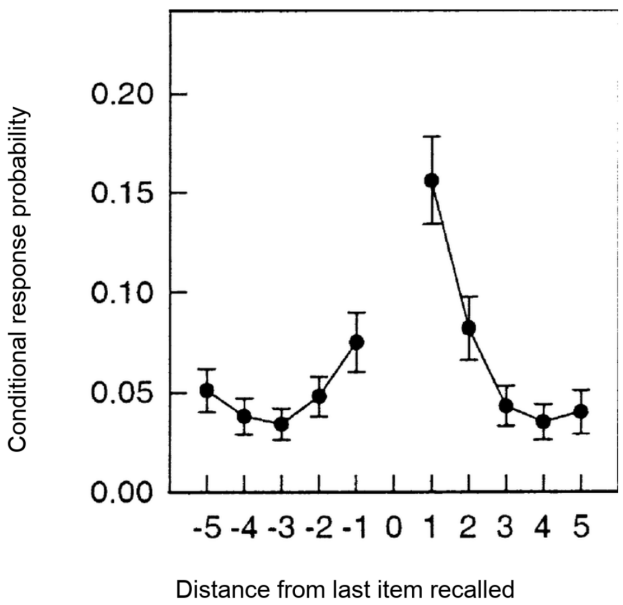
The average combines the word's occurrences in different sentence contexts.

Words joining nearer the labels have more similar patterns.

Elman (1990), Finding Structure in Time · Cognitive Science

Notes

Human recall favors nearby studied items



People study a list, then freely recall its items.

Horizontal axis: positions apart in the studied list.

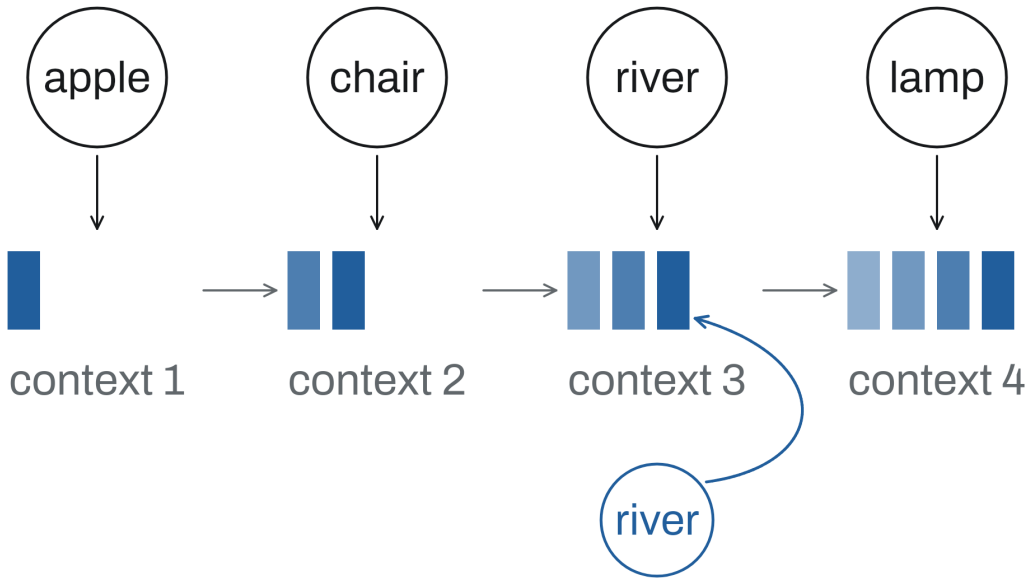
+1 means the item studied immediately after the one just recalled.

Nearby items are more likely to be recalled next.

Howard & Kahana (2002), Temporal Context Model · J. Mathematical Psychology

Notes

Temporal context explains nearby recall



Each bar group is a context vector; darker bars mean stronger components.

Nearby studied items acquire overlapping contexts.

Recall reinstates context, which cues nearby items.

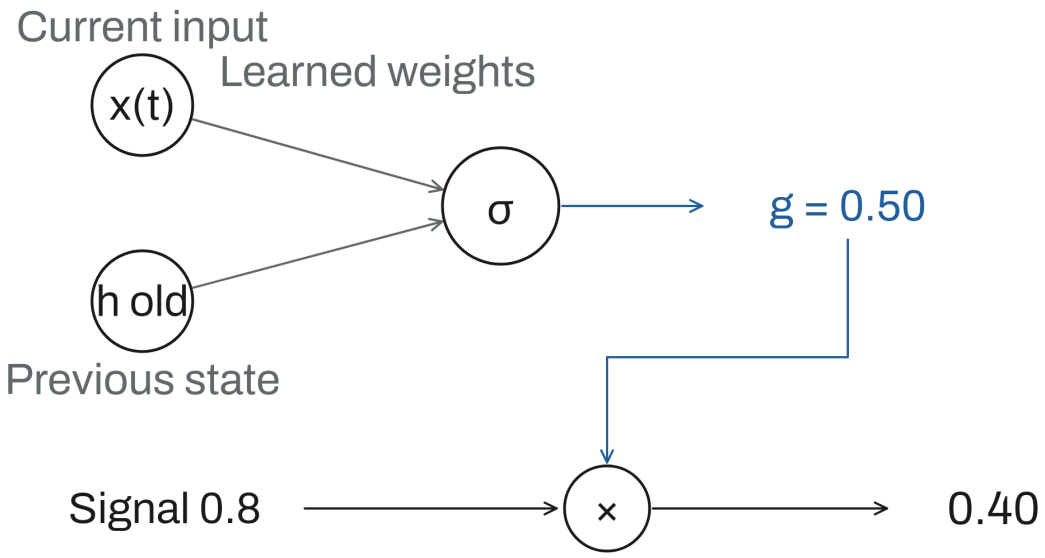
If recalling “river” brings back its context, which items gain a useful cue?

Items studied near “river” share more of that context.

Howard & Kahana (2002), Temporal Context Model · J. Mathematical Psychology

Notes

A gate is a learned amount



Current input and previous state determine the gate.

A sigmoid turns their weighted sum into an amount from 0 to 1.

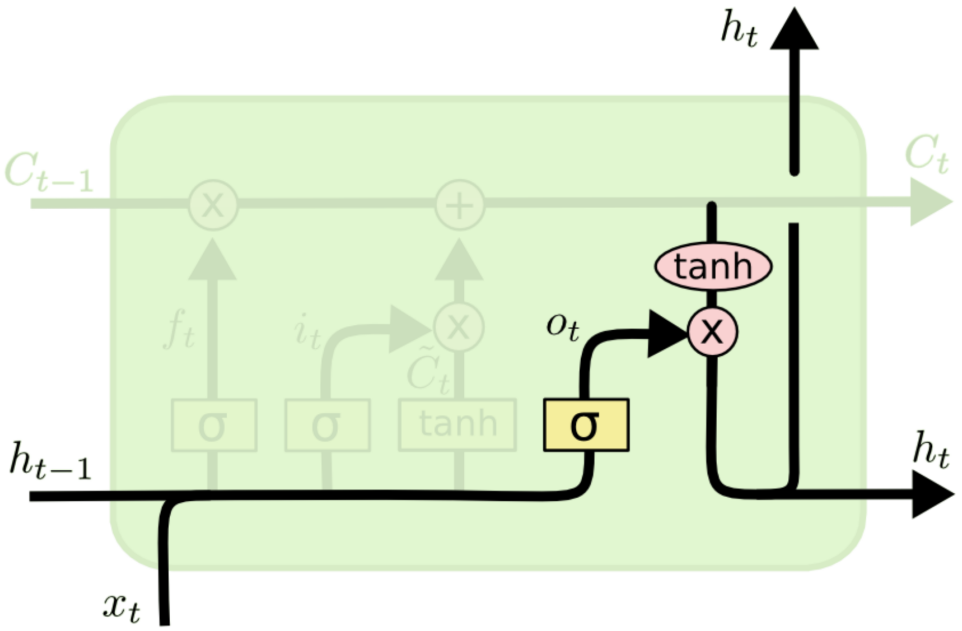
Multiplication scales how much a signal contributes.

Gate g: 0.50

Hand-set gate for demonstration

Notes

Long short-term memory (LSTM)



The cell state carries information forward.

The forget gate controls what to retain.

The input gate controls what to write.

Combine retained information with the new contribution.

The output gate controls the visible hidden state.

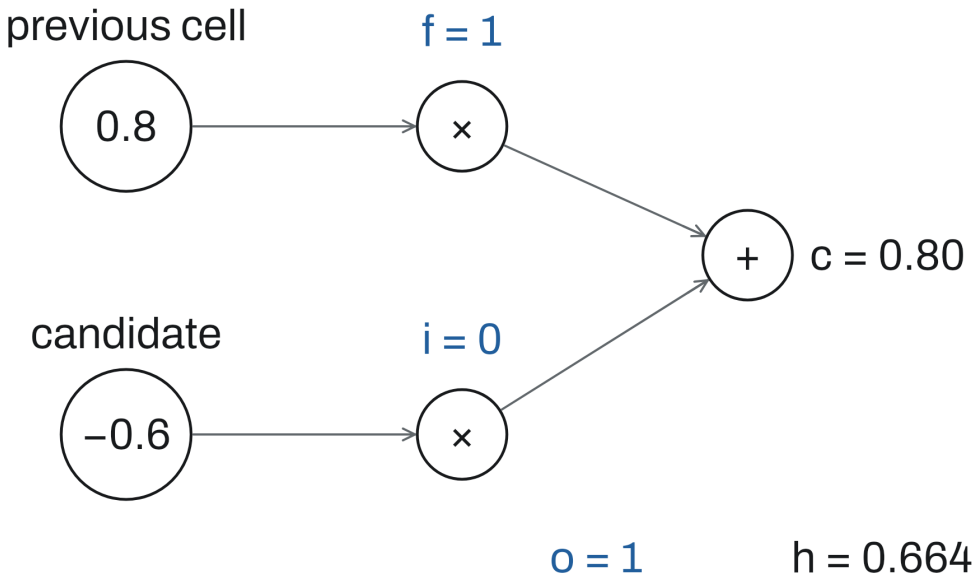
Can a stored cell value differ from the visible hidden output?

Yes. The output gate controls how much of tanh(cell) is exposed.

Christopher Olah (2015) · Diagram; Hochreiter & Schmidhuber (1997); Gers et al. (2000)

Notes

Preserve a value, then replace it



$c = f \times \text{previous } c + i \times \text{candidate}$

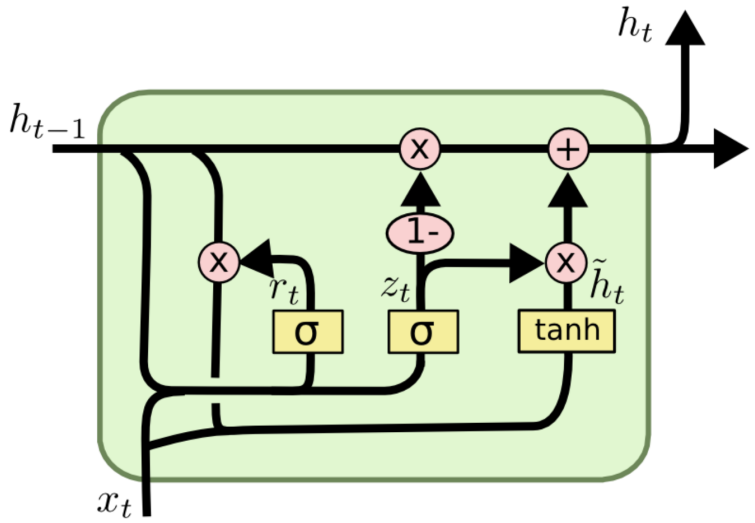
$h = o \times \tanh(c)$

$c = 0.800$

$h = 0.664$

Notes

Gated recurrent unit (GRU)



Old state 0.80 → new state 0.42
Candidate 0.04 · update gate writes 50%

One recurrent state, with no separate cell state.

The update gate controls the old/new blend.

Reset changes how the old state forms the candidate.

Update z (write): 0.50

Reset r: 1.00

Set writing to 0. Does changing reset change the stored state?

No. With writing at 0, the old state is retained regardless of the candidate.

Christopher Olah (2015) · Diagram; Cho et al. (2014)

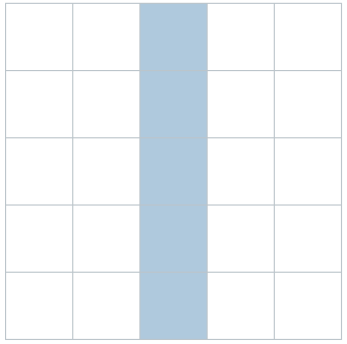
Notes

Same bird, different location

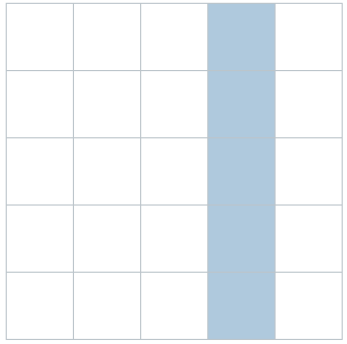
Notes

A visual pattern can move

Here



And here



The local pattern is the same.

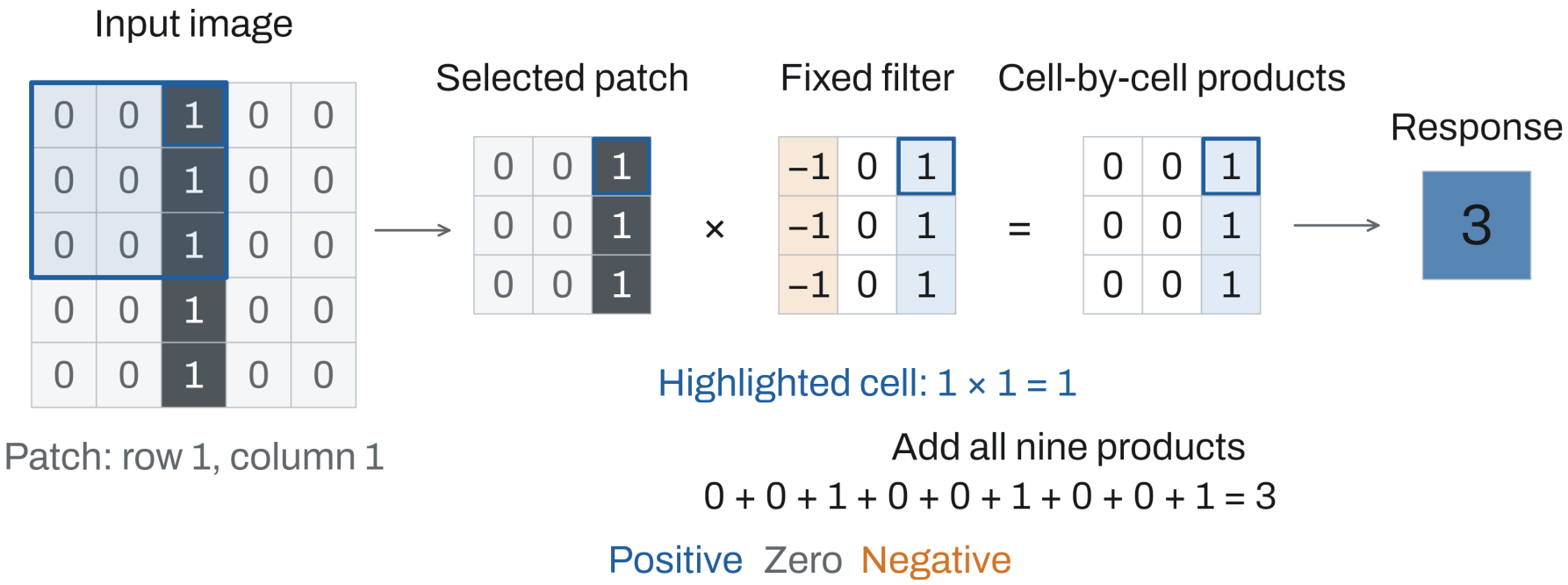
The same edge can appear in different places.

A dense network gives each location separate weights.

A convolution reuses a local detector.

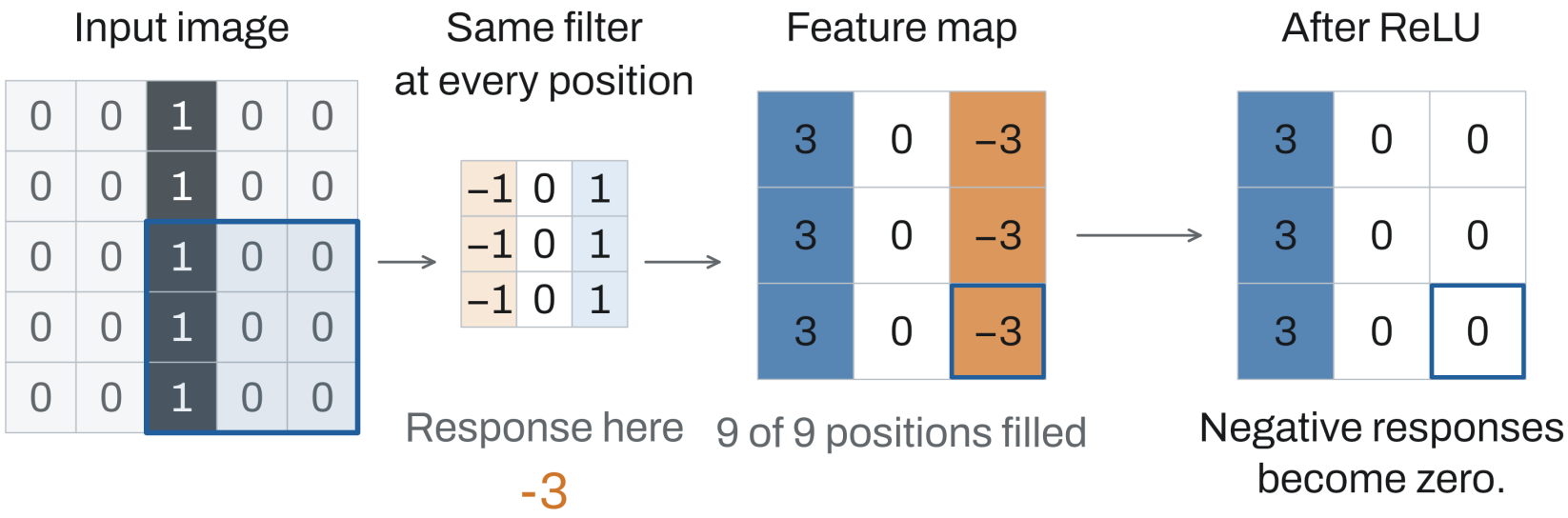
Notes

One patch produces one number



Notes

Build a feature map one position at a time

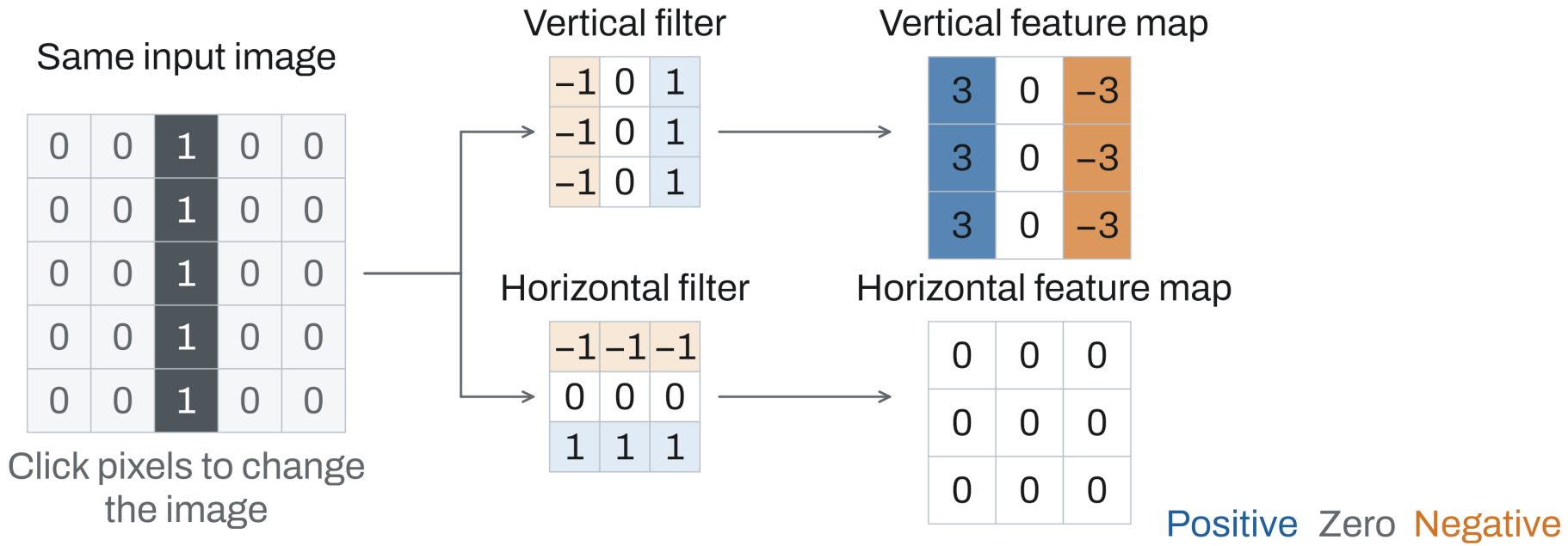


Each move fills one corresponding output cell.

Positive Zero Negative

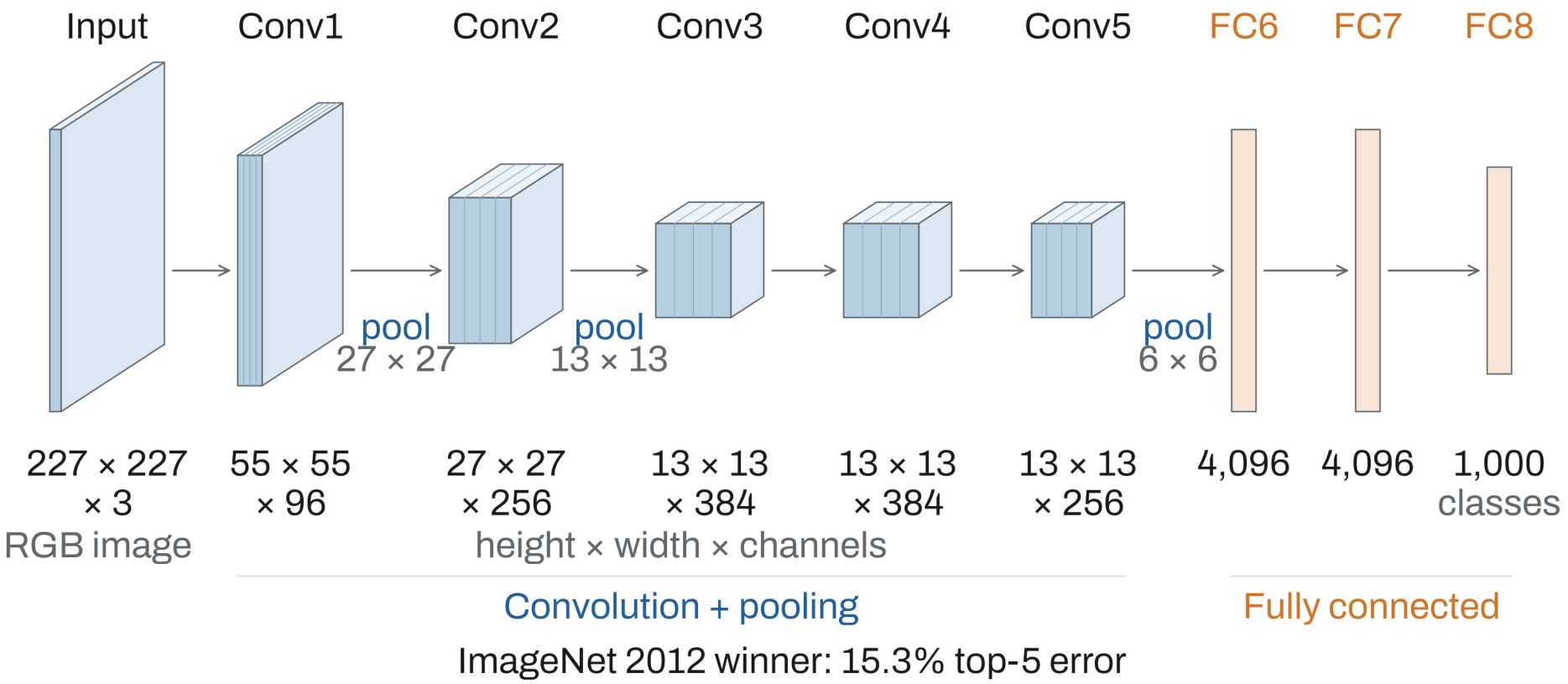
Notes

Two filters, two different feature maps



Notes

AlexNet: a convolutional neural network

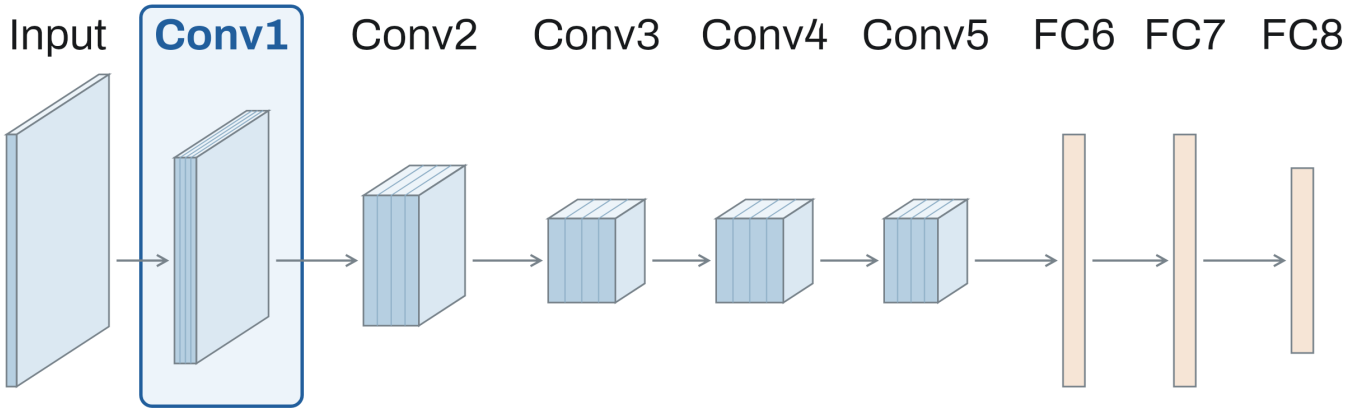


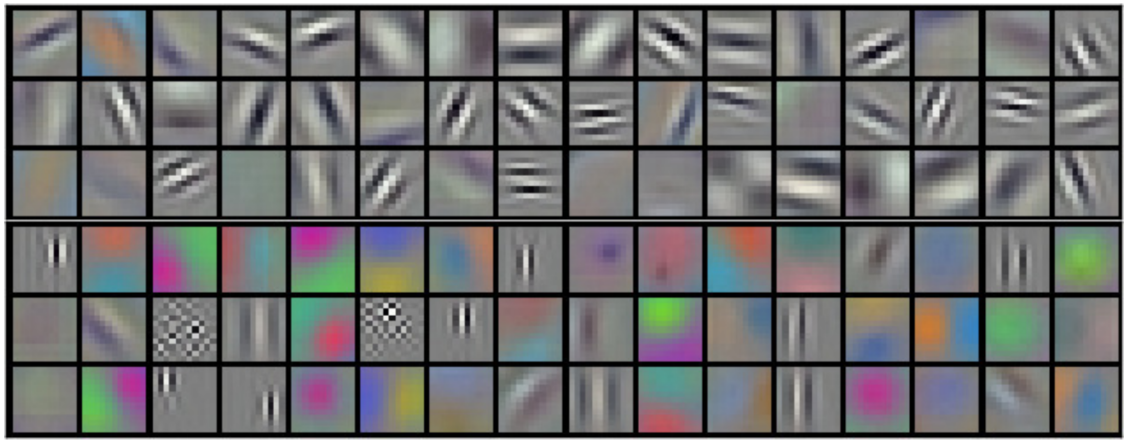
ImageNet 2012 winner: 15.3% top-5 error

Krizhevsky, Sutskever & Hinton (2012) · ImageNet classification with deep CNNs

Notes

AlexNet: learned Conv1 filters

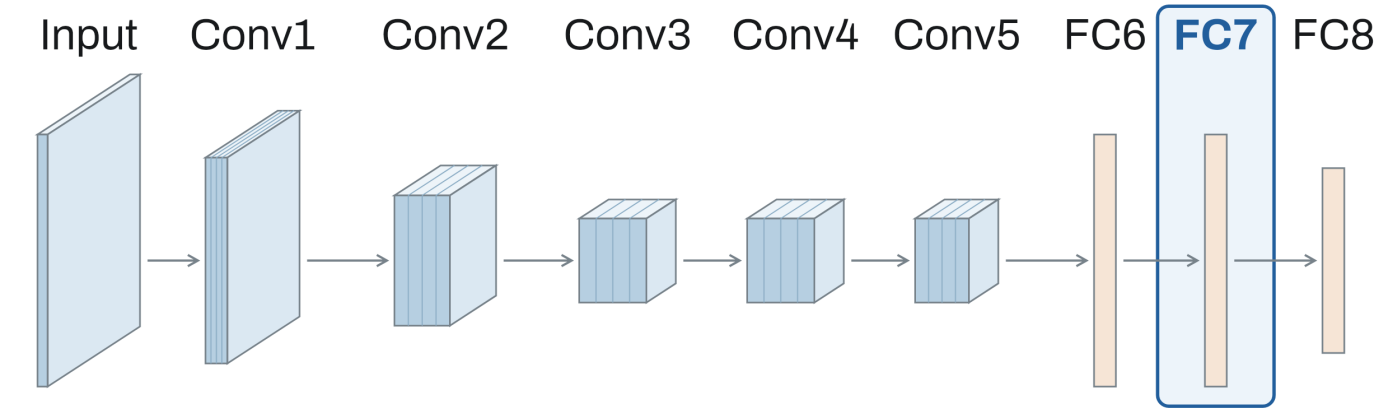




Krizhevsky, Sutskever & Hinton (2012) · Fig. 3

Notes

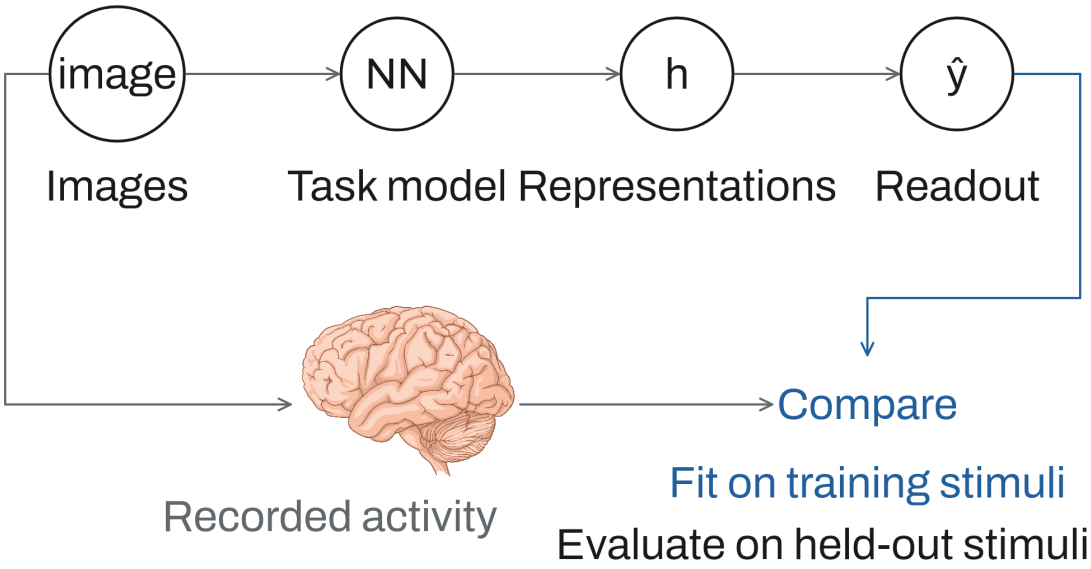
AlexNet: FC7 representations



Krizhevsky, Sutskever & Hinton (2012) · Fig. 4, right (cropped)

Notes

Compare a visual model with cortex



Optimize the model for object recognition.

Fit a readout to predict the recorded neural responses.

Compare predictions with recordings for held-out images.

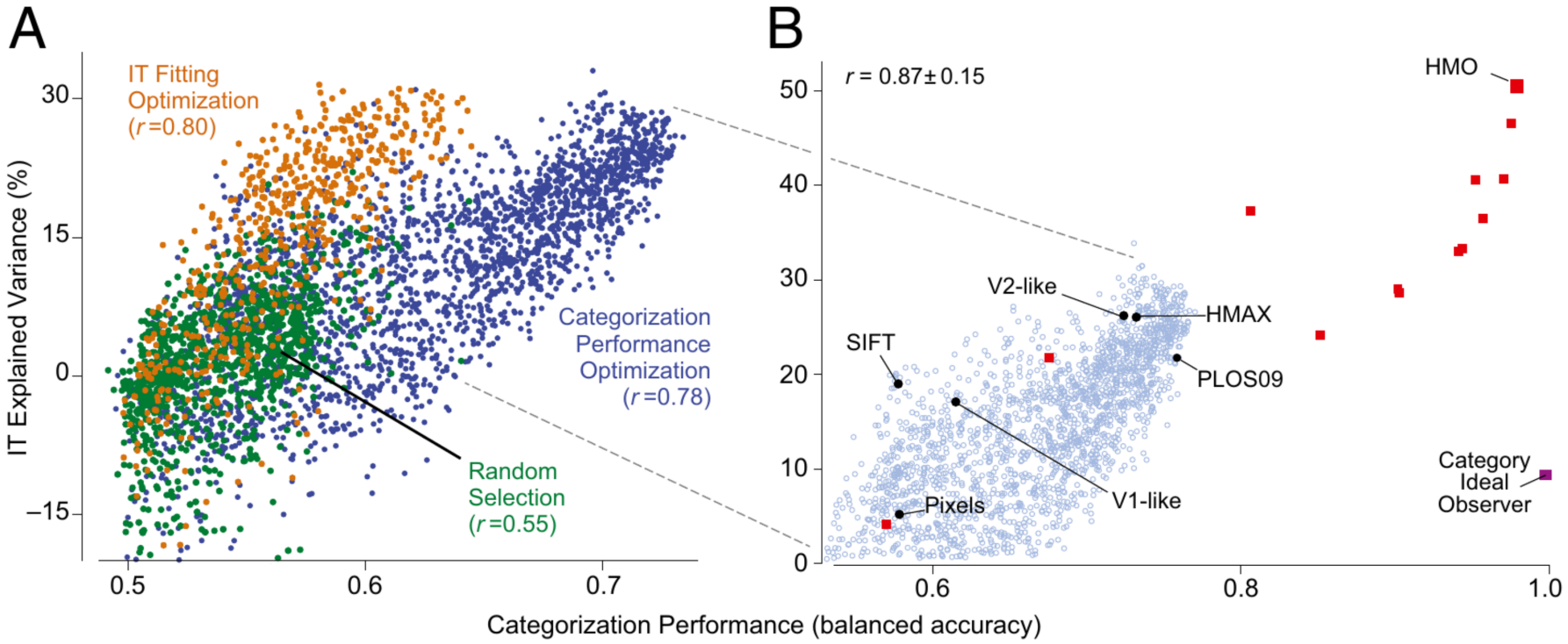
Why test the readout on different images?

To test generalization instead of measuring how well it fits the training images.

Yamins et al. (2014) · PNAS · doi:10.1073/pnas.1403112111

Notes

Task performance and IT predictivity



Yamins et al. (2014) · PNAS · doi:10.1073/pnas.1403112111

Notes

What the visual-cortex result supports

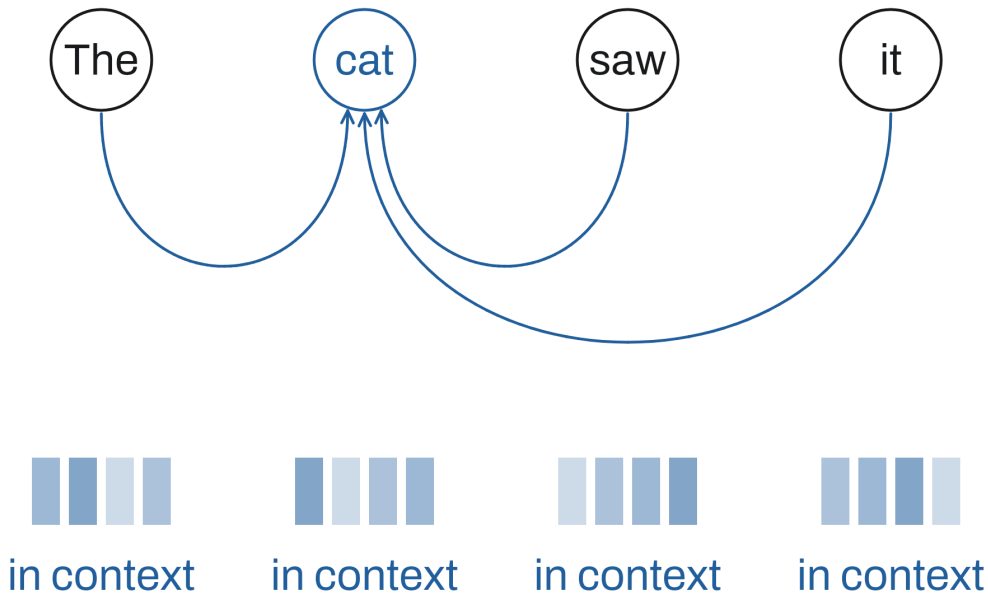
In these models, stronger object recognition tended to accompany better IT predictions.

Task optimization can help identify useful representations.

Yamins et al. (2014) · PNAS · doi:10.1073/pnas.1403112111

Notes

Transformers: represent a word using its context



Each word starts with a vector.

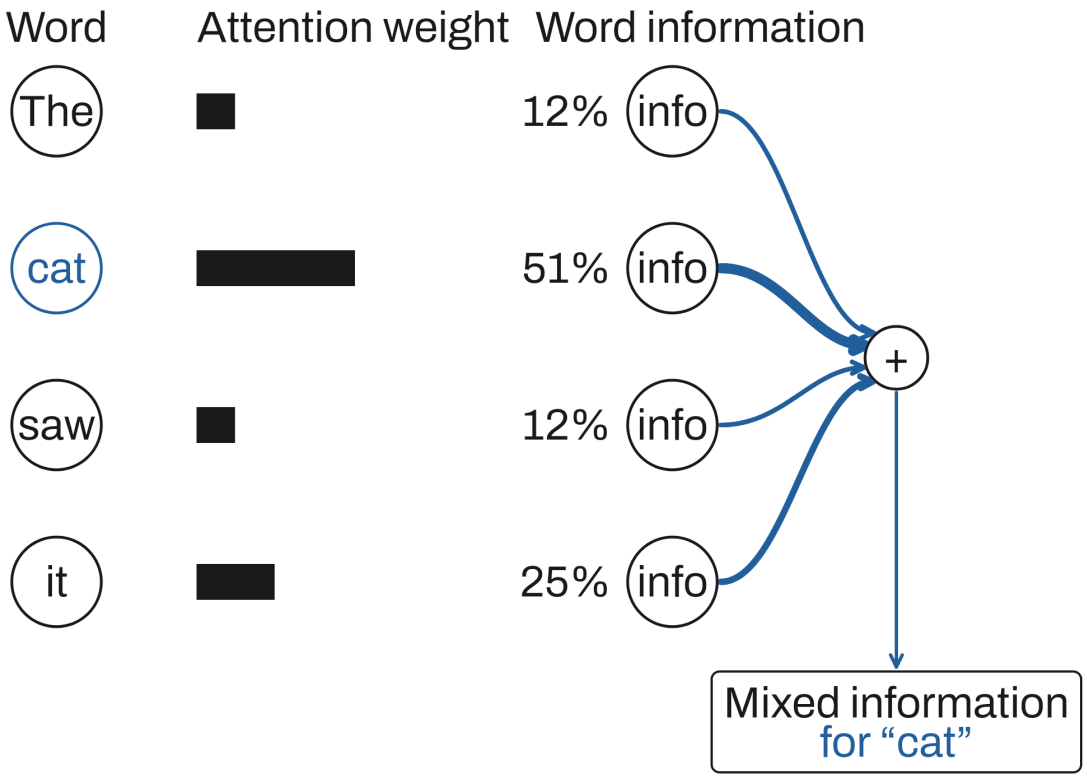
Bidirectional attention uses words before and after it.

Positions can be processed in parallel.

Vaswani et al. (2017); Devlin et al. (2019)

Notes

Attention weights control the information flow



Bidirectional example · illustrative weights

Each word can use context from both directions.

Multiply each word’s information by its attention weight.

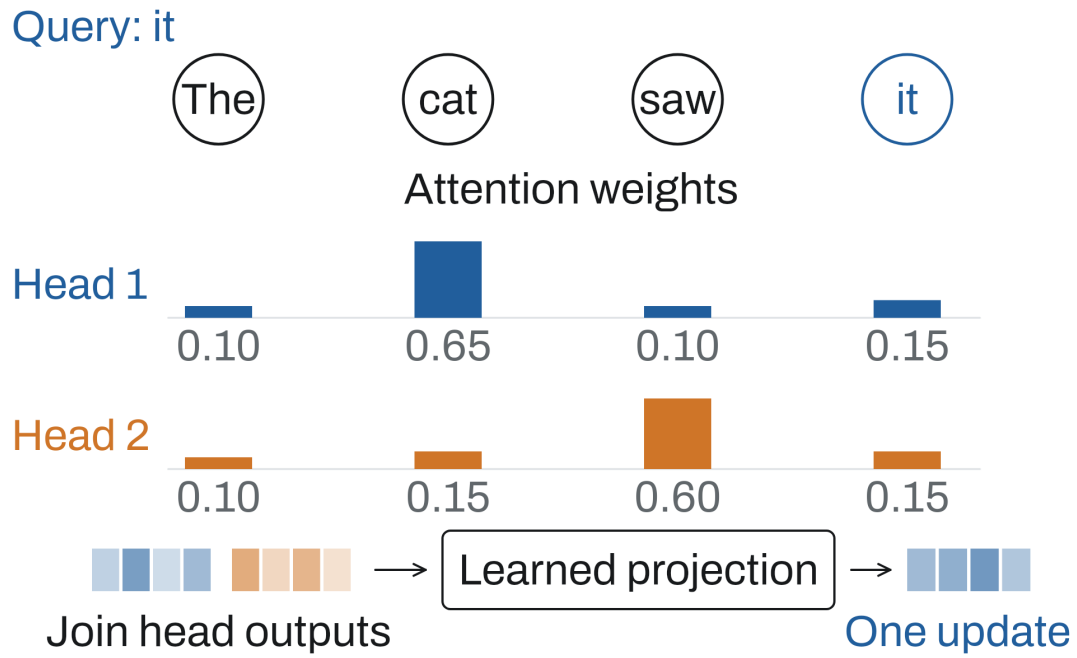
Add the weighted information to form an attention update.

Word being updated: cat

Illustrative weights

Notes

Several heads, several learned mixtures



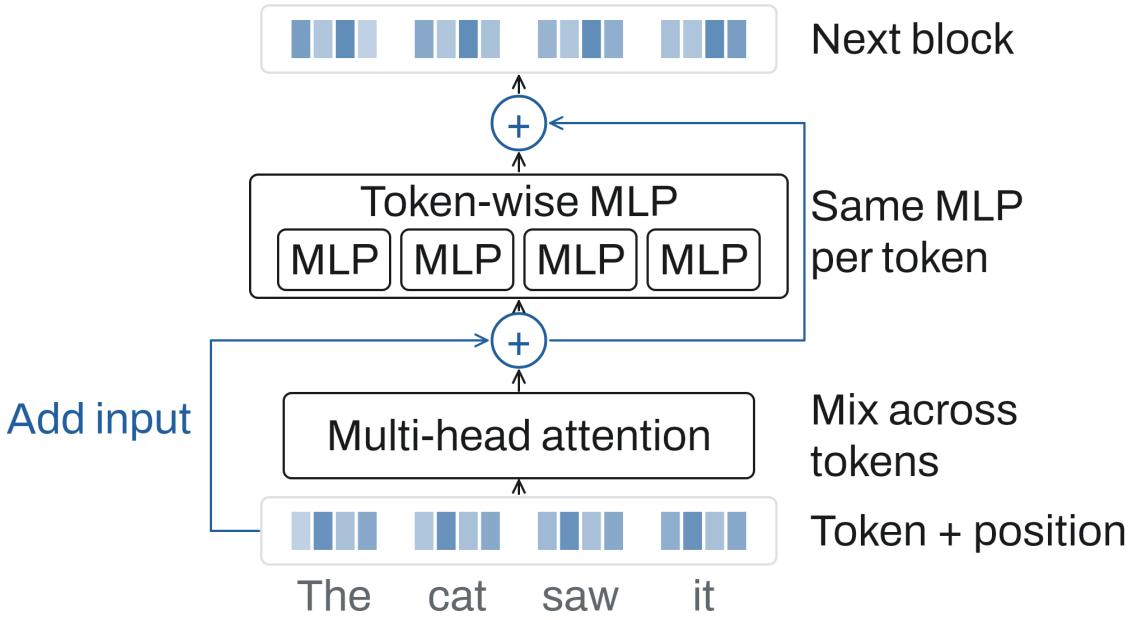
Different heads can use different attention patterns.

Their outputs are combined.

Head roles are learned rather than assigned in advance.

Notes

Attention is one part of a block



Attention mixes information across available positions.

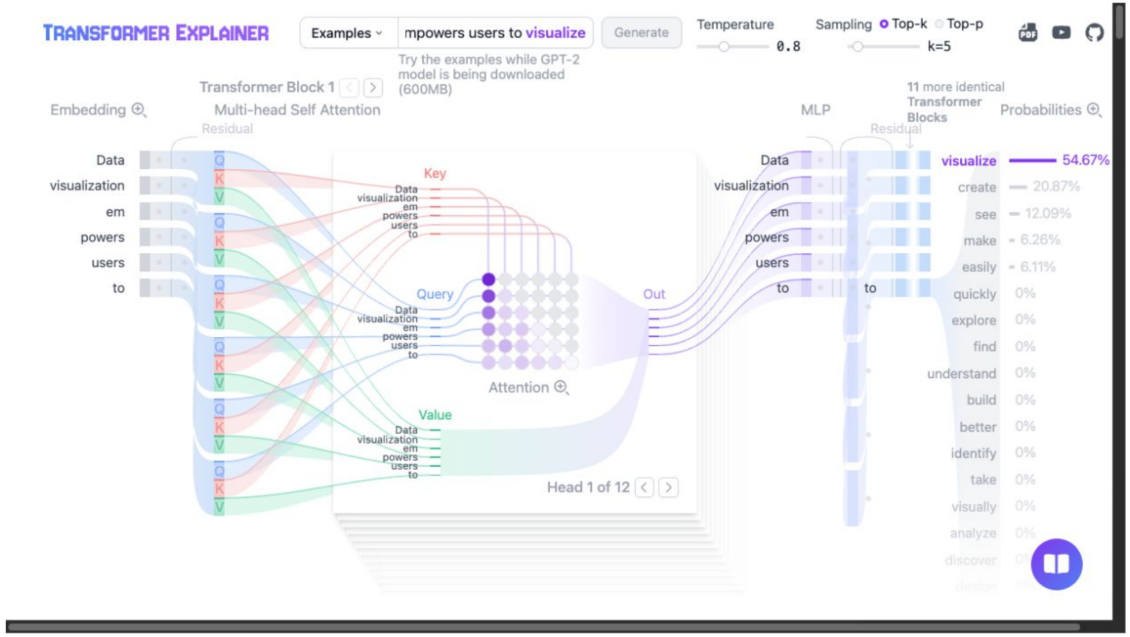
A feedforward network transforms each position.

A residual path keeps the previous representation and adds an update.

Vaswani et al. (2017); Radford et al. (2019)

Notes

Transformer demos



Transformer Explainer

GPT-2 · current and earlier words

3Blue1Brown

Attention, step by step · video

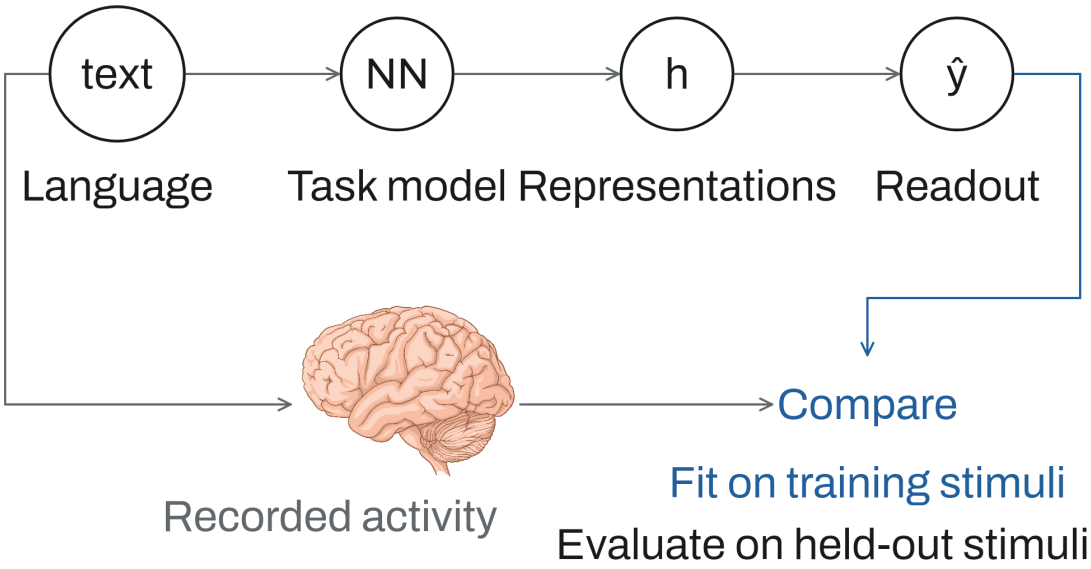
BertViz

BERT examples · both directions

Transformer Explainer · Georgia Tech Polo Club

Notes

Compare language models with brain responses



Present the same language to model and participants.

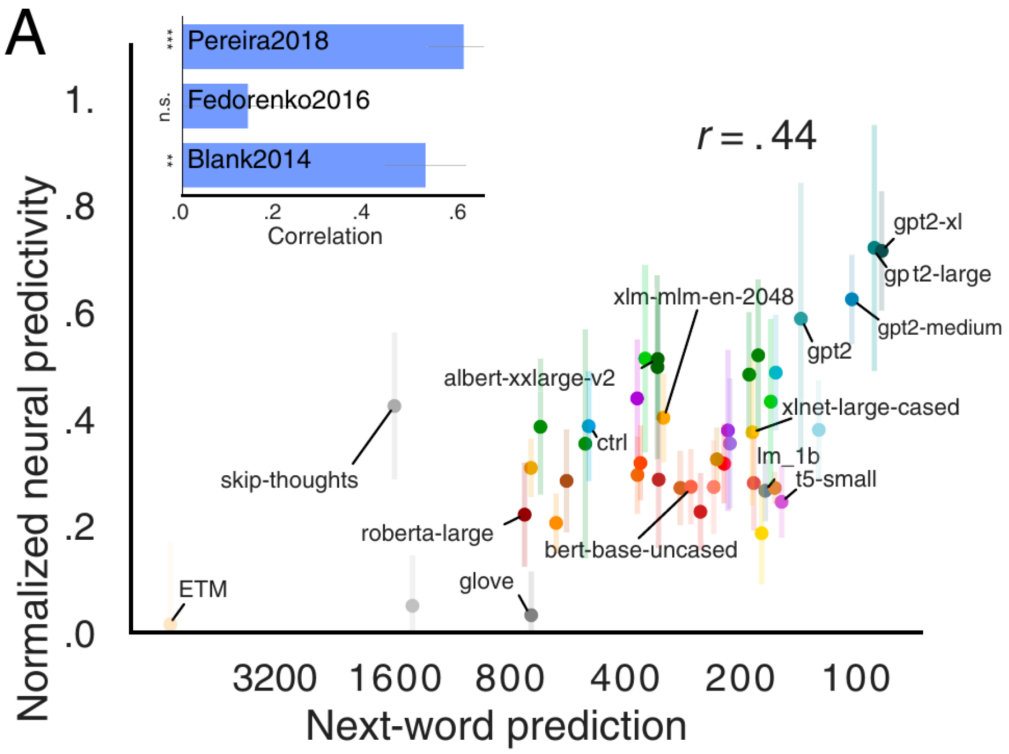
Fit a readout from representations to predicted responses.

Compare predictions with recordings for held-out stimuli.

Schrimpf et al. (2021) · PNAS · doi:10.1073/pnas.2105646118

Notes

Next-word performance and brain predictivity



Perplexity scores next-word prediction: lower is better. It decreases from left to right.

Higher on the vertical axis means better predictions of recorded brain responses.

Each point represents a model.

Schrimpf et al. (2021) · PNAS · doi:10.1073/pnas.2105646118

Notes

A predictive model is a testable hypothesis

Better next-word prediction was associated with better neural fits.

Architecture and training both mattered.

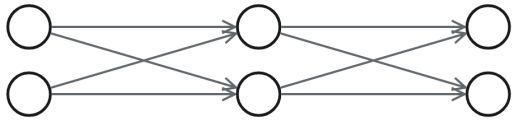
A good fit does not prove the brain implements GPT.

Schrimpf et al. (2021) · PNAS · doi:10.1073/pnas.2105646118

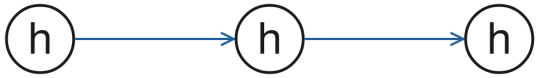
Notes

Choose the information flow

Layers



Recurrence



Convolution

Attention

MLP: combine input features.

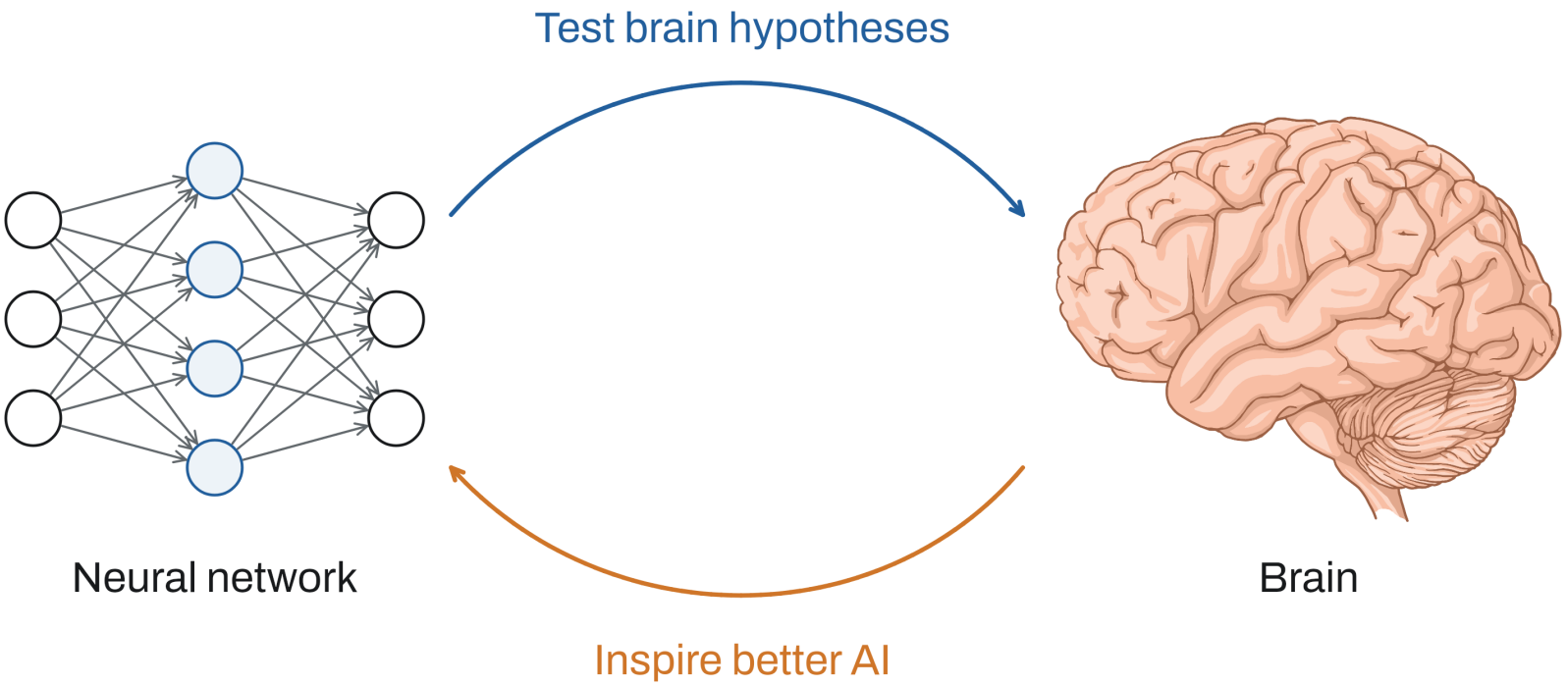
RNN / LSTM / GRU: update a state through time.

CNN: reuse local filters across space.

Transformer: mix available context with attention.

Notes

Neural networks as model organisms



Notes

References

Christopher Olah (2015) · Understanding LSTM Networks

Rogers & McClelland (2004) · Semantic Cognition

Saffran, Aslin & Newport (1996) · Statistical learning in infants

Elman (1990) · Finding Structure in Time

Howard & Kahana (2002) · Temporal Context Model

Hochreiter & Schmidhuber (1997) · LSTM

Gers, Schmidhuber & Cummins (2000) · Learning to Forget

Cho et al. (2014) · RNN Encoder–Decoder / GRU

Yamins et al. (2014) · Visual cortex

Vaswani et al. (2017) · Attention Is All You Need

Radford et al. (2019) · GPT-2

Schrimpf et al. (2021) · Language and brain

Devlin et al. (2019) · BERT

Mandler & McDonough (1993) · Concept Formation in Infancy

Rogers et al. (2004) · Semantic dementia

Anderson (2026) · Beni greenlet news report

Krizhevsky, Sutskever & Hinton (2012) · AlexNet

NIAID Visual & Medical Arts (2024) · Human brain illustration · Courtesy of NIAID

Notes